

But what IS a black hole?



Submission to SoME 2025

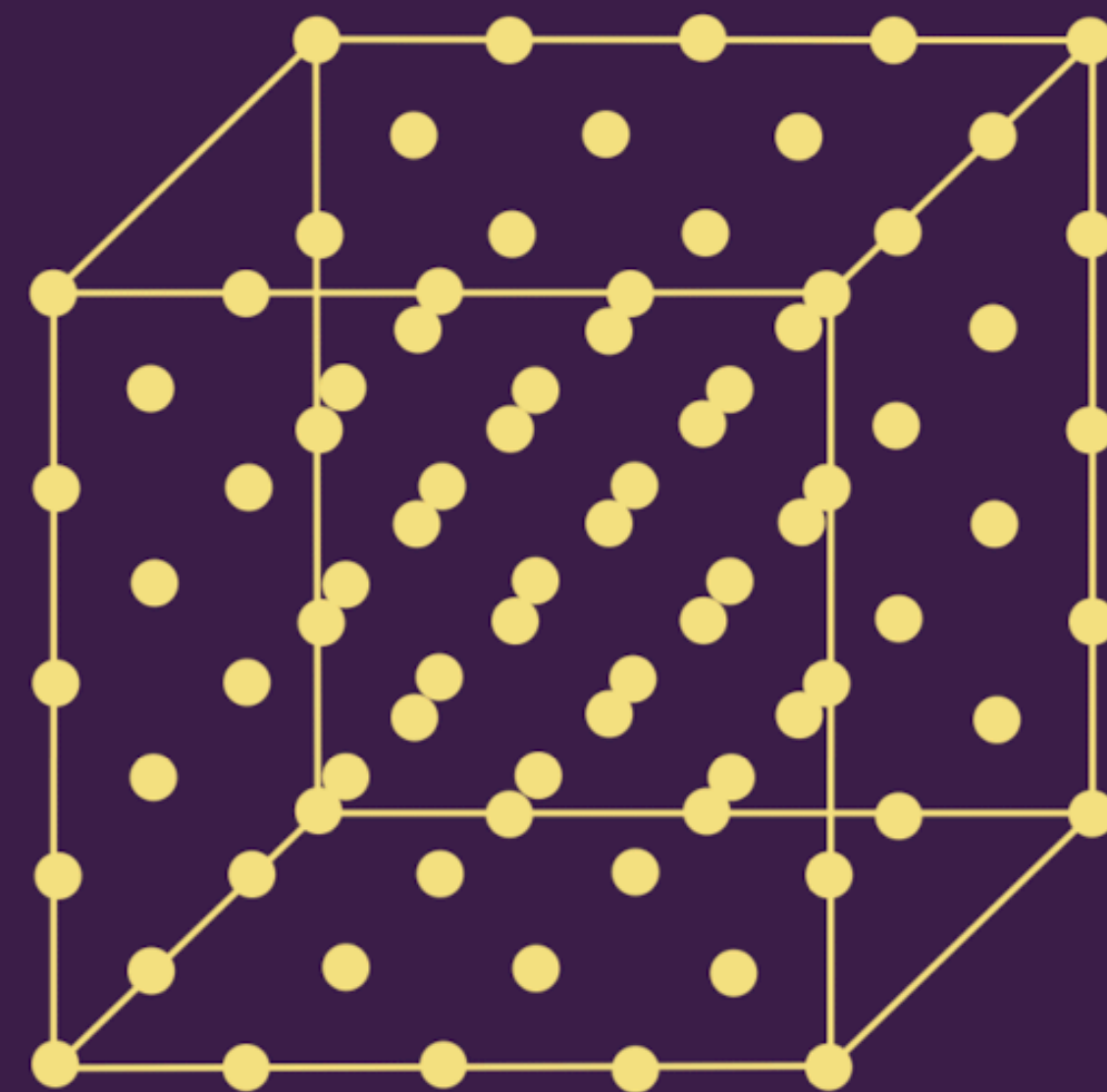
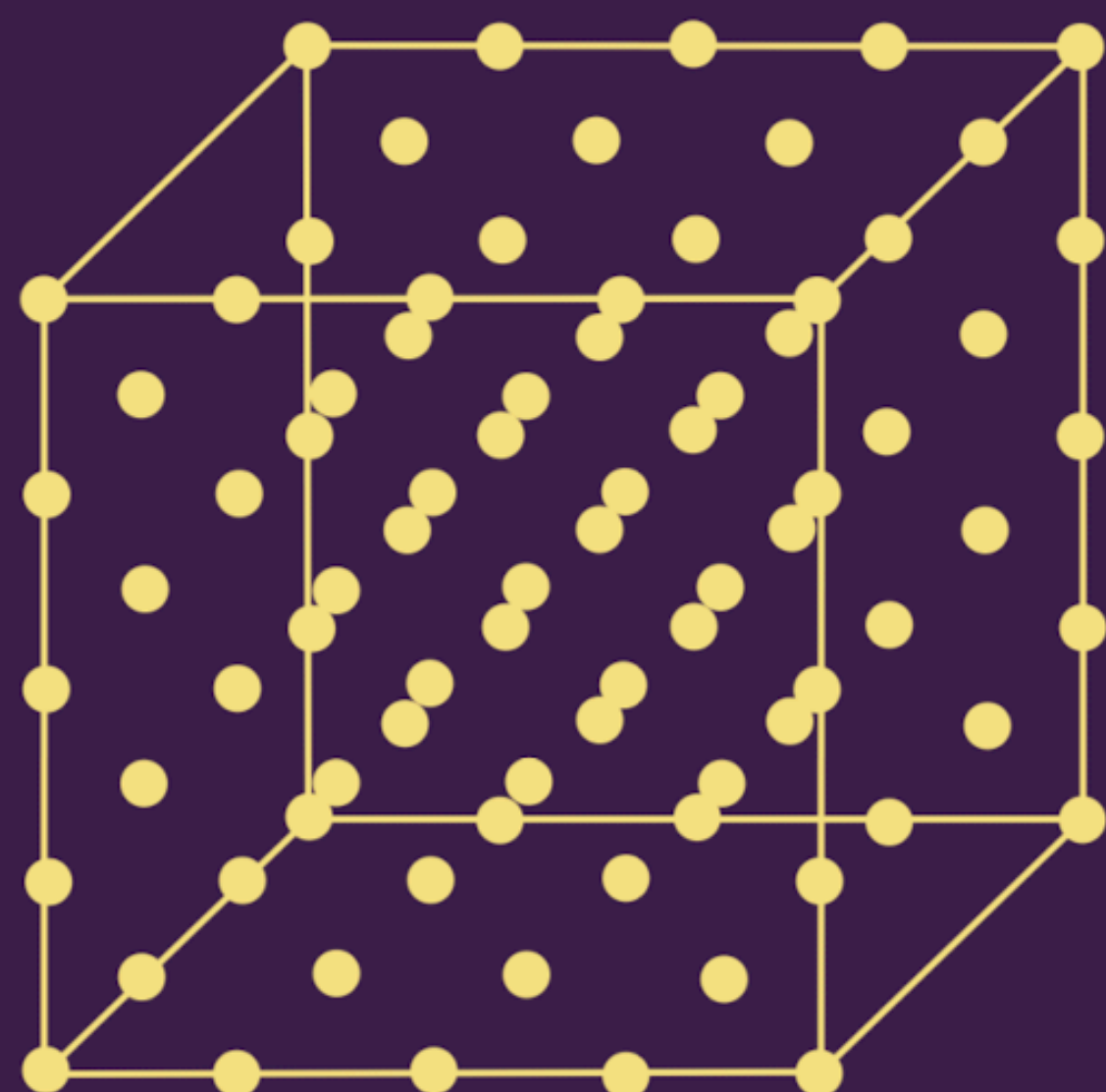
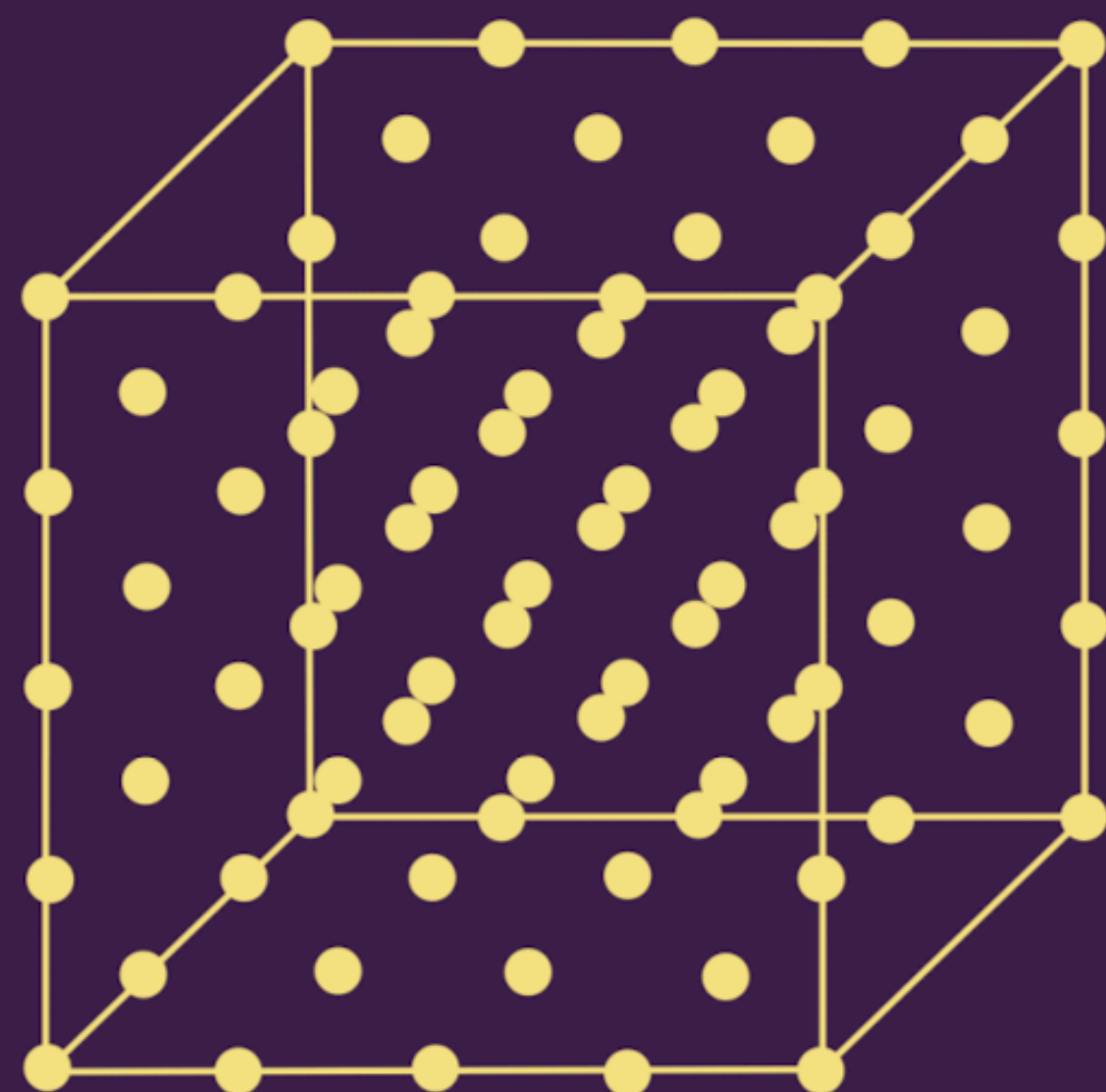
In the public domain, black holes are generally understood as local astrophysical phenomena. This short article aims to move beyond this treatment and impress upon an enthusiastic (non-expert) reader a global understanding of black holes. By exploring relativity via causal structure rather than tensor calculus, our discussion covers advanced topics whilst remaining accessible.

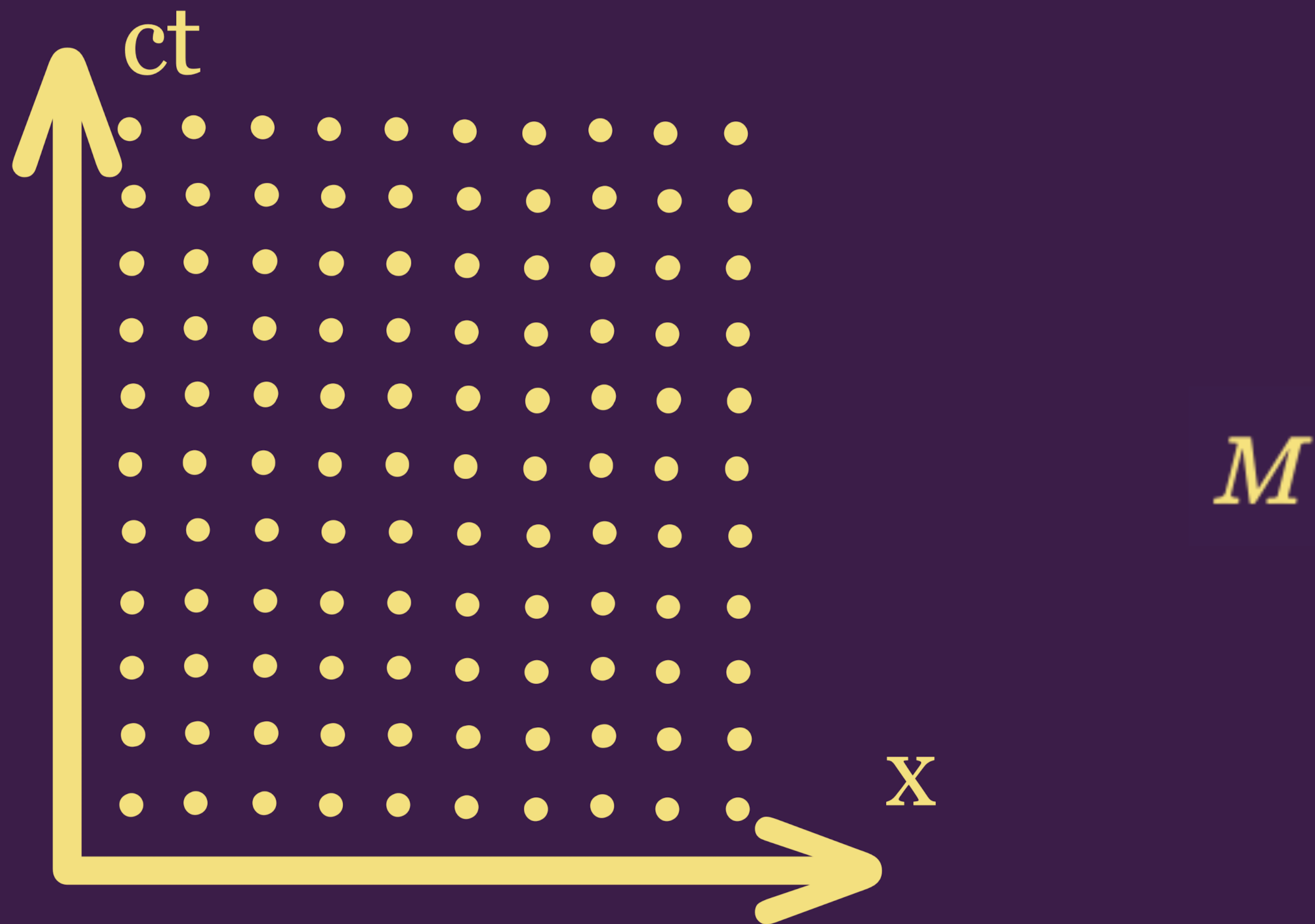


$$B = M / [M \cap J^-(\mathcal{I}^+)]$$

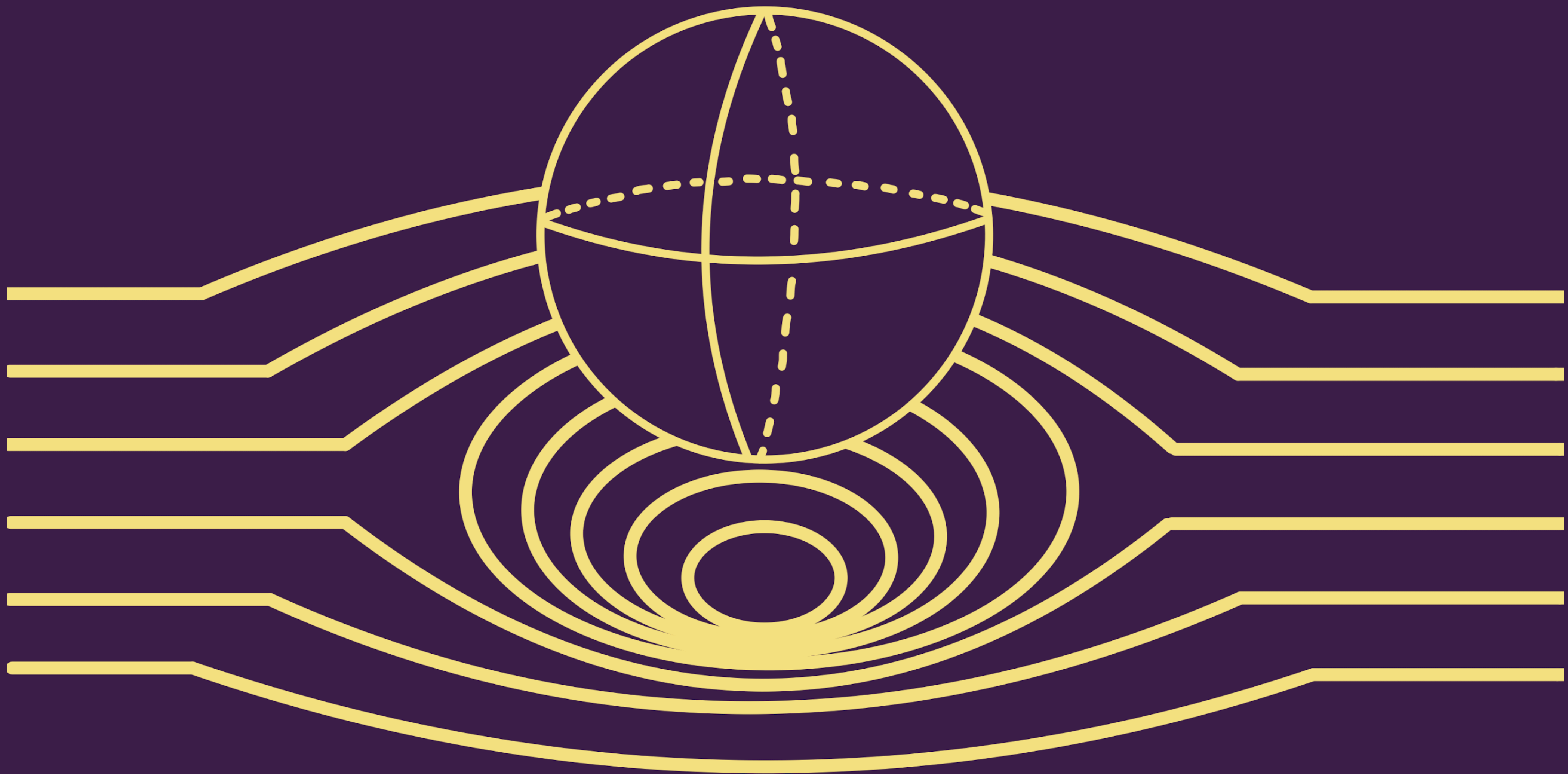
BEHOLD! A black hole.

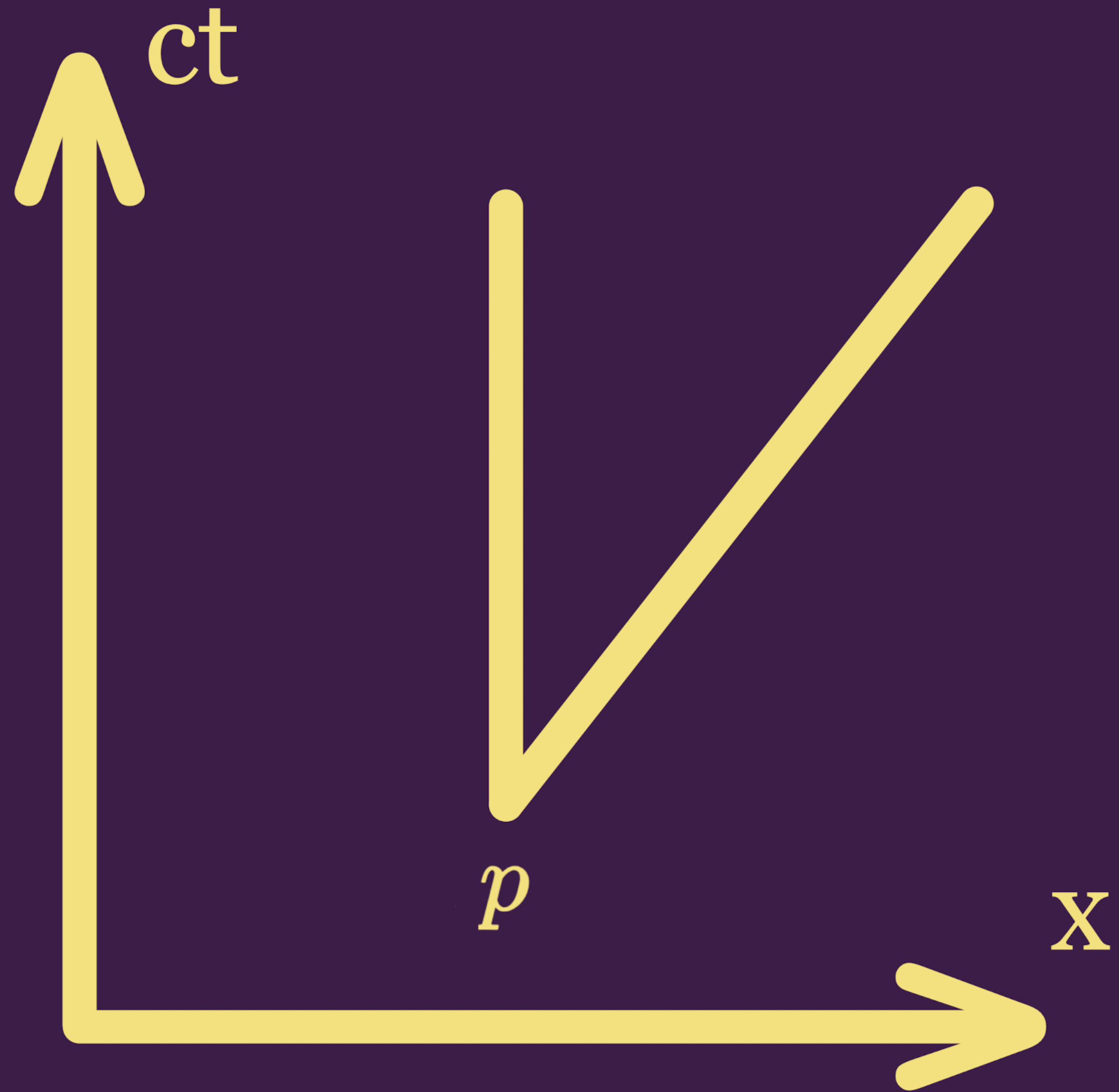
A black hole is a causally disconnected region of spacetime

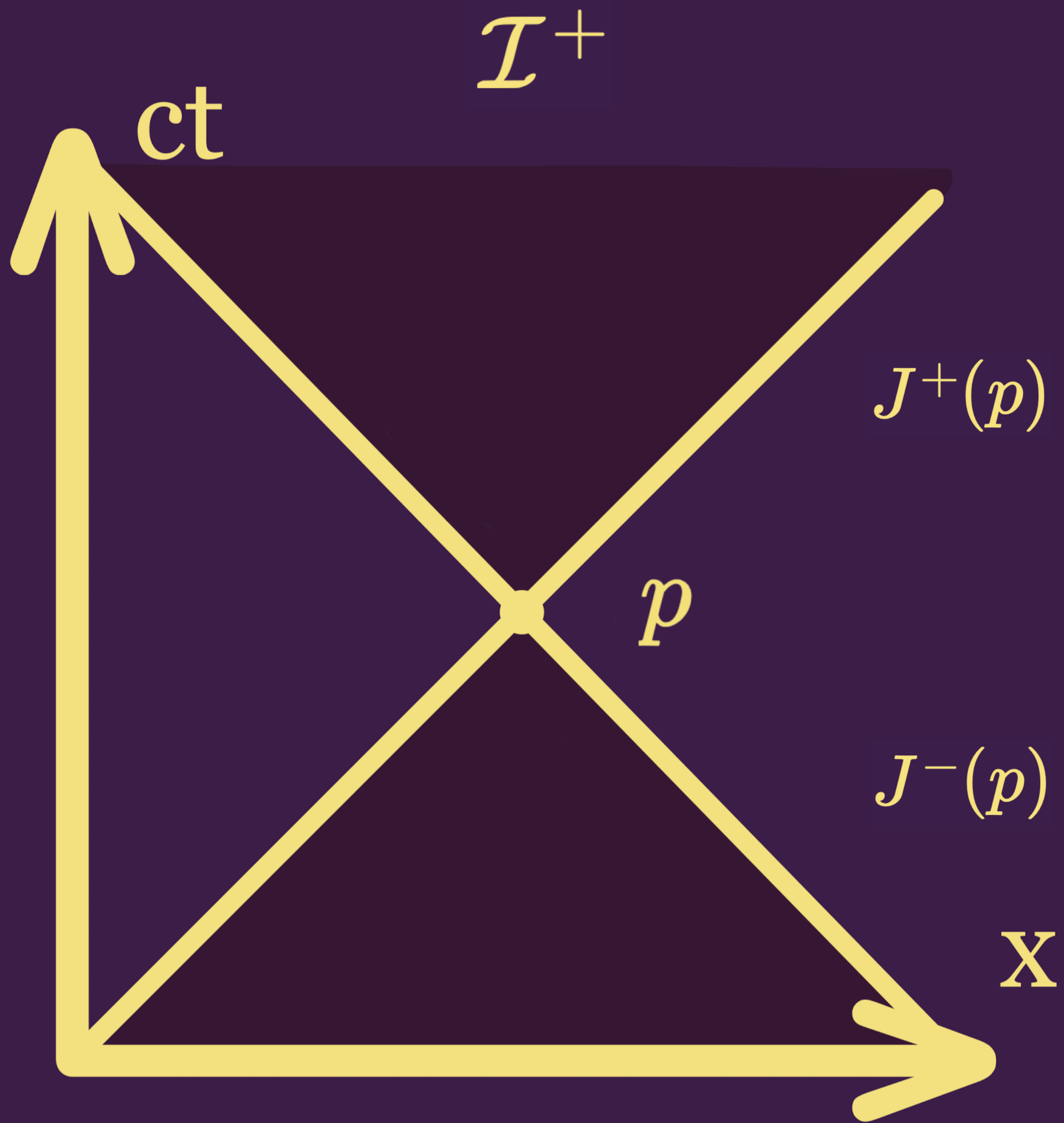




Spacetime tells matter how to move; matter tells spacetime how to curve







$$M \cap J^-(\mathcal{I}^+) = M \text{ ?}$$

